

DAK GURU STUDY MATERIAL

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DIGIPIN

Digital Postal Index Number

National Level Addressing Grid • Department of Posts, India

Comprehensive Examination Notes

LDCE — Inspector of Posts | PS Group 'B' | Departmental Examinations

Verified against the Final DIGIPIN Technical Document, DHRUVA Policy & PIB Launch (27 May 2025)

DISCLAIMER

This booklet has been prepared strictly from an examination point of view. While every effort has been made to ensure accuracy and the inclusion of the most recent changes, amendments, orders, and official documents, this document is intended as a study aid only. DIGIPIN and the DHRUVA framework constitute an evolving Digital Public Infrastructure; technical parameters, the institutional framework, and the legal regime may be revised by the Department of Posts. Readers and students are strongly advised to refer to official Government sources — the India Post website (indiapost.gov.in/digipin), PIB releases, the DIGIPIN Technical Document, the DHRUVA policy document, and the official open-source repository — for the most reliable and authoritative information.

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1. Introduction & Overview

DIGIPIN (Digital Postal Index Number) is a standardized, geo-coded, interoperable, grid-based **digital addressing system** for India, established by the **Department of Posts, Ministry of Communications**. It is an **open-source** national-level addressing grid developed in collaboration with **IIT Hyderabad** and the **National Remote Sensing Centre (NRSC), ISRO**.

The initiative seeks to provide simplified addressing solutions for the seamless delivery of public and private services and to enable “**Address as a Service**” (**AaaS**) across the country. DIGIPIN serves as a uniform address-referencing framework that is available both **offline and online**, bridging the gap between a physical location and its digital representation. The advent of DIGIPIN is described by the Department as a revolutionary step in India’s journey towards digital transformation.

THE ONE-LINE DEFINITION

DIGIPIN converts the **latitude and longitude** of any location in India into a unique, permanent, **10-character alphanumeric code** that represents an approximately **4-metre × 4-metre** square on the ground.

► 1.1 The Big Picture: Where DIGIPIN Fits

India has built a celebrated stack of Digital Public Infrastructure (DPI): **Aadhaar** for identity, **UPI** for payments, and **DigiLocker** for documents. The one missing layer was a **standardised digital address**. DIGIPIN — as the foundation of the **DHRUVA** framework — fills exactly that gap. The analogy most often used is:

Domain	DPI Layer	“Handle” Example
Identity	Aadhaar	12-digit number
Payments	UPI	name@bank (VPA)
Address	DHRUVA / DIGIPIN	name@dhruva

CRITICAL DISTINCTION (Exam-Important)

Do **not** equate DIGIPIN with the entire DPI. **DIGIPIN is the foundational addressing layer** (the grid). The broader DPI **framework** built on top of it is **DHRUVA** (Digital Hub for Reference & Unique Virtual Address). In short: DHRUVA is the ecosystem; DIGIPIN is the core grid that powers it.

2. Genesis & Key Milestones

DIGIPIN did not appear overnight. It evolved from an approach paper into a beta consultation and finally a published standard. The timeline below captures every milestone relevant to the examination.

Date / Period	Milestone
October 2021	Department of Posts released an approach paper proposing a “ Digital Address Code (DAC) ” — the conceptual origin of DIGIPIN, highlighting the need for an address identity linked to geospatial coordinates.
19 July 2024	Beta version of the DIGIPIN technical document released for public comments and expert opinion.
Up to 22 Sept 2024	Public feedback / stakeholder consultation window on the beta version.
March 2025	Final version of the DIGIPIN Technical Document released, establishing DIGIPIN as the foundation layer of the national addressing DPI.
27 May 2025	Official launch of the ‘Know Your DIGIPIN’ and ‘Know Your PIN Code’ web portals by the Department of Posts (PIB, Ministry of Communications). The DHRUVA policy document was released around the same time.
August 2025	MoU signed between the Department of Posts and MapmyIndia (Mappls) to integrate DIGIPIN into indigenous mapping platforms.

🕒 MOST-ASKED DATE

If only one date is asked, it is **27 May 2025** — the official launch of the DIGIPIN portal. Remember the supporting pair: **beta in July 2024, final technical document in March 2025, portal launch on 27 May 2025**.

3. Core Concept & Design Principles

DIGIPIN is visualised as an **alphanumeric, offline grid system** that divides the entire geographical territory of India into uniform **4-metre × 4-metre (approximate)** units. The seven defining principles are:

Principle	What It Means
Geo-coded	Each 4m × 4m unit gets a unique 10-character code derived solely from its latitude and longitude.
Privacy by design	The code represents only a location — no name, address, mobile, or email is attached. It can be openly published without privacy concerns.

Principle	What It Means
Permanent	The code does not change with names of states/cities/localities/roads, nor with new buildings or road-network changes.
Offline-capable	A GNSS-enabled device can compute the DIGIPIN without internet ; once generated it can be shared and used offline.
Directional	The logical labelling makes nearby cells share similar prefixes, giving an inherent sense of direction and adjacency .
Deterministic & reversible	Encoding (coordinates → code) and decoding (code → coordinates) are exact mathematical functions — no database lookup is needed.
Complementary	DIGIPIN does not replace the postal address or PIN Code; it is an additional digital layer of precision.

► 3.1 Coordinate Reference System (CRS)

The CRS used in the DIGIPIN design is **EPSG:4326**, also known as **WGS 84** (World Geodetic System 1984, datum at epoch 2005). This was chosen for its wide recognition and adoption, simplicity, and global coverage. EPSG:4326 is the same coordinate system used by GPS devices worldwide, which is what allows any GNSS receiver to feed directly into the DIGIPIN computation.

4. Code Architecture & The 16 Symbols

A DIGIPIN is an encoding of latitude and longitude into a sequence of alphanumeric symbols. The system uses exactly **16 symbols**:

2 3 4 5 6 7 8 9 C J K L M P F T

WHY THESE 16 SYMBOLS? (Frequently Asked)

The digits **0 and 1** are **deliberately excluded** because they are easily confused with the letters **O and I**. The remaining symbols were chosen to maintain **phonetic and visual clarity** and to avoid characters that look or sound alike. This makes the code easy to read aloud and record — a real benefit for rural and remote users.

► 4.1 Evolution from the Beta Version

The **final** version of DIGIPIN evolved from its beta through two key modifications, based on public and expert feedback:

- 1. Character replacement:** The beta characters **G, W, and X** were replaced by **C, F, and T** respectively, to enhance phonetic and visual clarity while preserving the overall sequence of characters.
- 2. Bounding-box change:** The bounding box was modified from the beta extent of **63.5°–99°E & 1.5°–39°N** to the final extent of **63.5°–99.5°E & 2.5°–38.5°N**.

► 4.2 The 10-Level Hierarchical Partition

Encoding is performed hierarchically across **10 levels**. Each level subdivides the previous region into a **4 × 4 grid of 16 sub-regions**, and one symbol is appended at each level:

- **Level-1:** A single bounding box covering the country is split into **16 (4×4) regions**. The chosen region's symbol becomes the **first character**.
- **Level-2:** Each Level-1 region is subdivided into 16 sub-regions. The first **two** characters thus identify one of **16² = 256** sub-regions.
- **Levels 3–10:** The same process repeats identically, each level adding one character and multiplying precision by 16.

The final **10-symbol** code uniquely identifies one of **16¹⁰** cells — over **1.09 trillion** (1,099,511,627,776) cells — within the national bounding box.

► 4.3 How a DIGIPIN Is Written (Display Format)

⚠ **FORMAT CORRECTION (Per the Official Repository)**

The **canonical** DIGIPIN is a **continuous 10-character string with no punctuation** (e.g. **39J49LL8T4**). For databases, APIs, and system-to-system exchange, hyphens and other punctuation are **not permitted**.

For **display only**, spaces may optionally be used in a **3-4-3 grouping** to aid readability (e.g. **39J 49LL 8T4**). Note: older notes and some portals show a legacy **3-3-4 hyphenated** form (e.g. **39J-49L-L8T4**); treat the no-punctuation 10-character string as the authoritative form.

5. Bounding Box & Grid Specifications

The bounding box used for DIGIPIN covers the following extent:

Longitude	63.5° – 99.5° East
Latitude	2.5° – 38.5° North
Total span	36° × 36° (latitude × longitude)

CRS	EPSG:4326 (WGS 84)
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This area deliberately includes the maritime **Exclusive Economic Zone (EEZ)**, allowing DIGIPIN to address Indian assets at sea — such as oil rigs and potential future man-made islands. The maritime EEZ is computed assuming a **200 nautical mile** extent from the coastline.

Why These Exact Corner Points?

- It includes the **entire territory of India**, plus the maritime EEZ.
- The grid aligns with the **Survey of India's** topographical-sheet (toposheet) numbering system.
- **Level-1 grid lines do not cut through** cities with very large populations.
- The Indian **mainland is covered by only 8 regions**, which can therefore be labelled with the digits 2–9 at Level-1.

► 5.1 Grid Size at Each Level

The table relates DIGIPIN code length to grid size (degrees and arc-minutes) and approximate linear distance **at the equator**. The base span is 36° ; each level divides each side by 4, so the angular size at level n is $36^\circ \div 4^n$.

Code Length	Grid (degrees)	Grid (arc-min)	Linear Distance	Scale
1	9°	540'	≈ 1000 km	Region
2	2.25°	135'	≈ 250 km	State
3	0.5625°	33.75'	≈ 62.5 km	District
4	0.1406°	8.44'	≈ 15.6 km	City
5	0.0352°	2.11'	≈ 3.9 km	Town
6	0.0088°	0.53'	≈ 1 km	Locality
7	0.0022°	0.13'	≈ 250 m	Block
8	0.00055°	0.033'	≈ 61 m	Cluster
9	0.00014°	0.0082'	≈ 15 m	Building
10	0.000034°	0.0021'	≈ 3.8 m	Doorstep

⚠ IMPORTANT CORRECTION (Verified against the Technical Document)

Many DIGIPIN notes in circulation contain a **garbled grid-size table** — e.g. showing Level-1 as “9” (arc-minutes) instead of $9^\circ (= 540')$, or Level-9 as “0.5”. These are wrong. The correct rule is simple: Level-1 = $9^\circ \times 9^\circ$, and every later level divides the side by 4. The figures above match the official document (Level-1 ≈ 1000 km, Level-6 ≈ 1 km, Level-10 ≈ 3.8 m). The “Scale” column is an intuitive aid, not an official label.

Rule of thumb: Code length 6 locates a $\sim 1 \text{ km} \times 1 \text{ km}$ region; the full **10-character** code locates a $\sim 3.8 \text{ m} \times 3.8 \text{ m}$ unit. At the equator the level-10 cell is about $3.8 \text{ m} \times 4 \text{ m}$, and its exact dimensions vary marginally with latitude (cells become slightly narrower in the east-west direction as one moves north).

6. The Labelling Grid & Symbol Assignment

⚠ MAJOR CORRECTION (Verified)

Some notes describe the assignment as a “Z-order / Peano curve starting from the bottom-left (South-West) corner.” This is **incorrect** for DIGIPIN. The official technical document specifies that, for **Levels 2–10**, symbols are assigned in an **anticlockwise fashion, spiralling outwards**, and that **Level-1 uses a different labelling grid** from Levels 2–10. Use the corrected description and the exact grid below.

► 6.1 Two Different Grids

- 1. Level-1 grid:** Designed so that the Indian **mainland is labelled using only the numerals 2–9** (the mainland falls within only 8 of the 16 regions). Using a distinct Level-1 grid avoids inadvertently labelling locations **outside** India with the reserved letters and keeps mainland codes numeric.
- 2. Levels 2–10 grid:** A single, consistent labelled grid is applied **identically** across all of Levels 2 to 10. Here the symbols are assigned in an **anticlockwise, outward-spiralling** pattern. This grid is **different** from the Level-1 grid.

► 6.2 The Exact Levels 2–10 Grid (4×4)

Derived from the official open-source encode/decode logic, the Levels 2–10 labelling grid maps each of the 16 cells to a symbol as follows. North is at the top, South at the bottom; West is on the left, East on the right:

N ↑ / W ←	West	← →	← →	East
North	F	C	9	8
↑	J	3	2	7
↓	K	4	5	6
South	L	M	P	T

🔍 HOW TO READ THIS GRID

At every level from 2 to 10, the system finds which of these 16 cells contains the point, appends that cell's symbol, and “zooms in” to that cell for the next level. Because consecutive symbols (e.g. **2 and 3**, or **6 and 7**) sit next to each other, locations with similar codes are **geographical neighbours** — this is the built-in directionality. (The grid was verified by decoding Dak Bhawan's official DIGIPIN back to within ~2 m of its published coordinates.)

► 6.3 Grid-Line Tie-Breaking Rule

Occasionally a point's coordinates fall exactly on a grid line. The convention is: if the point lies on a **vertical (north–south)** grid line, it is assigned the DIGIPIN of the **adjoining right-side (eastward)** grid cell.

7. How DIGIPIN Is Generated — The Algorithm

The Department of Posts has released the DIGIPIN logic in the **public domain** (open source, Apache License 2.0). The system exposes two operations — **encode** and **decode** — both of which are deterministic and require **no database**.

► 7.1 Encoding: Coordinates → DIGIPIN

1	Start with the full bounding box: latitude 2.5°–38.5°N, longitude 63.5°–99.5°E.
2	Divide the current box into a 4 × 4 grid of 16 equal sub-cells.
3	Find which sub-cell contains the point (latitude, longitude).
4	Append that sub-cell's symbol (from the labelling grid) to the code.

- 5 Replace the bounding box with that sub-cell and repeat from Step 2.
- 6 After 10 iterations, the 10-character DIGIPIN is complete.

► 7.2 Decoding: DIGIPIN → Coordinates

- 1 Remove any spaces/hyphens; confirm the code is exactly 10 valid characters.
- 2 Start with the full bounding box.
- 3 For each character, locate its cell in the 4×4 labelling grid and shrink the bounding box to that cell.
- 4 After all 10 characters, the centre of the final tiny box is the decoded latitude/longitude.

KEY INSIGHT

Because both directions are pure mathematics, DIGIPIN needs **no server, no database, and no internet** to work. This is what makes it genuinely **offline** and **privacy-preserving** — there is simply no table linking a code to a person. The accuracy of a generated DIGIPIN depends on the precision of the **GNSS device** used to capture the coordinates.

► 7.3 Access Points (Official)

Resource	Official Link
Know Your DIGIPIN	dac.indiapost.gov.in/mydigipin/home
Know Your PIN Code	dac.indiapost.gov.in/mypincode/home
Initiative & FAQ page	indiapost.gov.in/digipin
Open-source repository	github.com/INDIAPOST-gov/digipin (Apache 2.0)
Technical Document (PDF)	indiapost.gov.in → Offerings → Initiatives → DIGIPIN

REPOSITORY NOTE (Updated)

The DIGIPIN open-source code was originally hosted at github.com/CEPT-VZG/digipin but has been **officially relocated** to github.com/INDIAPOST-gov/digipin. The old repository is no longer maintained. Always cite the new official repository.

8. Worked Example — Dak Bhawan (Step by Step)

The official technical document illustrates DIGIPIN using **Dak Bhawan, New Delhi** (the headquarters of the Department of Posts) at coordinates **28.622788°N, 77.213033°E**. The resulting DIGIPIN is:

39J 49LL 8T4

(canonical form: 39J49LL8T4)

The table traces how each of the 10 characters is selected, progressively narrowing the location from a 1000-km region down to a single ~3.8-metre cell:

Level	Symbol	Latitude Range (N)	Longitude Range (E)
1	3	20.50 – 29.50	72.50 – 81.50
2	9	27.25 – 29.50	77.00 – 79.25
3	J	28.375 – 28.9375	77.00 – 77.5625
4	4	28.5156 – 28.6562	77.1406 – 77.2812
5	9	28.6211 – 28.6562	77.2109 – 77.2461
6	L	28.6211 – 28.6299	77.2109 – 77.2197
7	L	28.6211 – 28.6233	77.2109 – 77.2131
8	8	28.6227 – 28.6233	77.2126 – 77.2131
9	T	28.6227 – 28.6229	77.2130 – 77.2131
10	4	28.6228 – 28.6228	77.2130 – 77.2131

✓ SELF-VERIFIED

Reading the symbols top to bottom gives **3 → 9 → J → 4 → 9 → L → L → 8 → T → 4 = 39J49LL8T4**. Decoding this code back through the same grid returns the centre point **28.62279°N, 77.21305°E** — within about **2 metres** of the published coordinates, confirming the example and the labelling grid are accurate.

9. The DHRUVA Ecosystem

DHRUVA — Digital Hub for Reference & Unique Virtual Address — is the comprehensive digital-address ecosystem in which DIGIPIN acts as the **foundational layer**. The DHRUVA policy document was released by the Ministry of Communications around the May 2025 launch. It is designed as a **Digital Public Infrastructure (DPI)** for addressing, analogous to how **UPI** functions for payments. Its architecture is **privacy-by-design, consent-based, and federated**, governed by Indian law including the **Digital Personal Data Protection Act, 2023 (DPDPA)**.

► 9.1 Core Objectives

- **Standardisation:** A uniform addressing format across India.
- **Interoperability:** Digital addresses usable seamlessly across platforms (e-commerce, emergency services, logistics, governance).
- **Privacy by design:** Decoupling physical locations from personal identity, with consent-based sharing.
- **Inclusivity:** A formal, verifiable address for every citizen — including unplanned settlements, rural areas, forests, and remote regions.

► 9.2 Key Components (Institutional Framework)

The DHRUVA framework defines the following roles:

Component	Role
Central Mapper (CM)	The primary authority that manages the DIGIPIN grid and the allocation of address suffixes (AIP suffixes); maintains the translation between DIGIPIN and other address formats.
Authorized Address Validation Agency (AAVA)	Entities authorised to verify the authenticity of a physical address associated with a DIGIPIN / virtual address.
Address Information Provider (AIP)	Entities that provide or host address data within the ecosystem.
Address Information User (AIU)	Services — delivery apps, government portals, etc. — that consume the address data to fulfil a service.
Address Information Agent (AIA)	An agent role within DHRUVA that facilitates address-related interactions in the ecosystem.

► 9.3 User Features

- **Digital Address Creation:** Generate a unique digital address for a precise location.
- **Consent-based Data Sharing:** Residents control who can access their precise location data.
- **Secure Address Verification:** Helps prevent fraud and ensures accurate service delivery.
- **Virtual Address (handle):** A handle such as **dakguru@dhruva** that masks the underlying DIGIPIN or coordinates — conceptually similar to a **UPI ID**.

10. DIGIPIN vs PIN Code & Global Systems

Alongside DIGIPIN, the Department launched ‘**Know Your PIN Code**’ on the same day (27 May 2025). The six-digit **PIN Code**, introduced in **1972**, remains the backbone of postal delivery. To modernise it, the Department geo-fenced all PIN Code boundaries nationwide.

- Over **1.5 lakh** PIN Code areas have been geo-fenced.
- Using **GNSS**, users can find the correct PIN Code for their exact location and submit feedback.
- The dataset is published on the **Open Government Data Platform** as “All India PIN Code Boundary geo-json”.

► 10.1 PIN Code vs DIGIPIN

Feature	PIN Code (1972)	DIGIPIN (2025)
Precision	Broad locality / post-office area	~4 m × 4 m exact cell
Format	6 digits (numeric)	10 characters (alphanumeric)
Basis	Administrative zones	Latitude & longitude (geo-coded)
Personal data	Tied to delivery area	None — pure location
Works offline	N/A (lookup-based)	Yes (computed locally)
Changes over time	Can be re-organised	Permanent for a location
Role	Sorting & routing of mail	Precise digital addressing layer

► 10.2 How DIGIPIN Compares Globally

DIGIPIN belongs to a family of grid/geocoding systems. Unlike most proprietary alternatives, it is **government-backed, open-source, and privacy-first**, and is purpose-built for India's territory and toposheet system. Comparable systems include **Google Plus Codes (Open Location Code)**, **what3words**, and the **Universal Postal Union (UPU)** addressing initiatives. A distinguishing feature of DIGIPIN is that it requires **no licensing**, stores **no personal data**, and works fully **offline**.

11. Benefits, Applications & Significance

► 11.1 Sector-wise Benefits

Sector	Benefit
Disaster Management	Precise location identification for rescue where landmarks may be destroyed.
Logistics & E-commerce	Cuts “last-mile” delivery failures from vague addresses; aids couriers & platforms (Amazon, Flipkart).
Emergency Response	Faster police / ambulance / fire response by pinpointing exact spots.
BFSI / KYC	An additional address attribute that improves accuracy and efficiency of KYC processes.
Governance	Better utility planning and accurate beneficiary onboarding for government schemes; potential ABDM linkage.
Inclusivity	A formal address for rural areas, forests, informal settlements, and maritime EEZ zones.
Navigation	Codes can be integrated into navigation apps for precise tracking.

► 11.2 Policy Alignment

DIGIPIN aligns with the **National Geospatial Policy 2022**, which seeks to strengthen India's geospatial sector for national development, economic prosperity, and a thriving information economy. It was officially reviewed and adopted by the **Thematic Working Group on Address** under that policy, and adds to the country's geospatial knowledge stack. It complements related initiatives such as **NAKSHA** (urban land survey), **ULPIN** (Unique Land Parcel Identification Number) under **DILRMP**, and the wider India Stack (Aadhaar, UPI, DigiLocker, ONDC).

🔔 CURRENT STATUS (Per Official FAQ)

The official version of DIGIPIN has been released and the public web application is live. However, per the India Post FAQ, **citizens cannot yet quote a DIGIPIN for booking articles at a post office** — the Department will officially announce this once DIGIPIN is fully incorporated into its operational systems.

12. Privacy, Security & Limitations

► 12.1 Why DIGIPIN Is Private by Design

- DIGIPIN is a **pure function of latitude and longitude** — it stores no name, address, mobile, or email.
- Two different residents at the **same coordinates** get the **same** DIGIPIN — it is **not** a personal identifier. Floor or flat details do **not** change it.
- Because there is **no database** linking codes to people, publishing a DIGIPIN does not expose personal data.
- The DHRUVA layer adds **consent-based** sharing and **DPDPA-2023** governance for any service that links an address to an identity.

► 12.2 Honest Limitations (for balanced understanding)

- **GNSS dependence:** Accuracy of a generated DIGIPIN depends on the precision of the GNSS device; poor signal yields a less precise code.
- **Vertical ambiguity:** DIGIPIN is 2-D — it does **not** distinguish floors in a multi-storey building.
- **Adoption curve:** Benefits depend on widespread adoption by couriers, apps, and government systems, which is still ongoing.
- **Not yet operational for booking:** As noted, it is not yet usable for booking postal articles at counters.

13. Frequently Asked Questions (Official)

The following Q&A are drawn from the official India Post DIGIPIN FAQ and are commonly examined.

Q. Will DIGIPIN replace my postal address?

A. No. DIGIPIN complements traditional addresses, adding a digital layer of precision; it will not replace existing postal addresses.

Q. Is my personal data stored in DIGIPIN?

A. No. DIGIPIN only encodes a geographic location; no personal data is stored, ensuring privacy.

Q. Does DIGIPIN work without internet?

A. Yes. It is an offline system — once generated, a DIGIPIN can be shared and used without an internet connection.

Q. Can two homes in one building have different DIGIPINs?

A. No. DIGIPIN is a pure function of latitude/longitude; floor or resident details do not change it. It is an addressing reference, not a person identifier.

Q. Is DIGIPIN applicable only within India?

A. Yes. It is designed for India's geographical boundaries, including maritime regions, within the bounding box 63.5°–99.5°E and 2.5°–38.5°N.

Q. How do I find my DIGIPIN?

A. Use a GNSS-enabled device to get your latitude/longitude, then convert it via the web application at dac.indiapost.gov.in/mydigipin/home. No login or signup is required.

Q. How precise is DIGIPIN?

A. Each code represents an approx. 4 m × 4 m square. Real-world accuracy depends on the precision of the GNSS device used.

Q. Can I use DIGIPIN to book an article at a post office now?

A. Not yet. The Department will announce this once DIGIPIN is incorporated into its operational systems.

14. Important Abbreviations

Abbreviation	Full Form
AaaS	Address as a Service
AAVA	Authorized Address Validation Agency
ABDM	Ayushman Bharat Digital Mission
AIA	Address Information Agent
AIP	Address Information Provider
AIU	Address Information User
API	Application Programming Interface
CM	Central Mapper
CRS	Coordinate Reference System
DAC	Digital Address Code
DHRUVA	Digital Hub for Reference & Unique Virtual Address
DIGIPIN	Digital Postal Index Number
DILRMP	Digital India Land Records Modernization Programme
DPDPA	Digital Personal Data Protection Act, 2023
DPI	Digital Public Infrastructure
EEZ	Exclusive Economic Zone
EPSG	European Petroleum Survey Group (geodetic dataset)
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
IIT	Indian Institute of Technology
ISRO	Indian Space Research Organisation
KYC	Know Your Customer
NAKSHA	National Geospatial Knowledge-based Land Survey of Urban Habitations
NRSC	National Remote Sensing Centre
ONDC	Open Network for Digital Commerce
PIN	Postal Index Number
ULPIN	Unique Land Parcel Identification Number
UPI	Unified Payments Interface

Abbreviation	Full Form
UPU	Universal Postal Union
WGS 84	World Geodetic System 1984

15. Quick Revision Capsule

A one-glance summary of the most frequently examined facts.

Parameter	Value
Full form	Digital Postal Index Number (DIGIPIN)
Launched by	Department of Posts, Ministry of Communications
Developed with	IIT Hyderabad and NRSC, ISRO
Official launch	27 May 2025 (portals); final technical document March 2025
Beta released	19 July 2024 (feedback up to 22 Sept 2024)
Conceptual origin	Digital Address Code (DAC) approach paper, Oct 2021
Code length	10 characters (alphanumeric)
16 symbols	2 3 4 5 6 7 8 9 C J K L M P F T
Excluded characters	0 and 1 (confusable with O and I)
Beta → final swap	G, W, X replaced by C, F, T
Cell size	Approx. 4 m × 4 m (Level-10 ≈ 3.8 m at equator)
Total levels	10 (each a 4×4 = 16 sub-region split)
Total cells	16 ¹⁰ (≈ 1.09 trillion)
Bounding box	63.5°–99.5°E, 2.5°–38.5°N
CRS	EPSG:4326 (WGS 84)
Total span	36° × 36°
Level-1 region	9° × 9° (≈ 1000 km)
Level-6 ≈	1 km × 1 km
Labelling (L2–10)	Anticlockwise, spiralling outwards
Display format	Continuous 10 chars; optional 3-4-3 spacing
Maritime coverage	Includes EEZ (200 nautical miles from coast)
Ecosystem / DPI	DHRUVA (Digital Hub for Reference & Unique Virtual Address)

Parameter	Value
DHRUVA components	CM, AAVA, AIP, AIU, AIA
Privacy	Stores no personal data; pure function of lat/long
Governing law	Digital Personal Data Protection Act, 2023
Open-source repo	github.com/INDIAPOST-gov/digipin (Apache 2.0)
Worked example	Dak Bhawan (28.622788°N, 77.213033°E) → 39J49LL8T4
Companion launch	‘Know Your PIN Code’ (1.5 lakh+ PIN areas geo-fenced)

16. Practice Questions (Self-Test)

Test your recall. Answers follow the questions.

1. DIGIPIN was developed by the Department of Posts in collaboration with which two institutions?

Ans: IIT Hyderabad and NRSC, ISRO.

2. On what date were the ‘Know Your DIGIPIN’ and ‘Know Your PIN Code’ portals launched?

Ans: 27 May 2025.

3. How many symbols does DIGIPIN use, and why are 0 and 1 excluded?

Ans: 16 symbols; 0 and 1 are excluded as they are confusable with O and I.

4. What is the bounding box of the DIGIPIN grid?

Ans: Longitude 63.5°–99.5°E; Latitude 2.5°–38.5°N (CRS: EPSG:4326 / WGS 84).

5. What approximate ground area does a full 10-character DIGIPIN represent?

Ans: Approximately 3.8 m × 3.8 m (about 4 m × 4 m).

6. How are symbols assigned across Levels 2–10?

Ans: In an anticlockwise, outward-spiralling pattern (Level-1 uses a different grid).

7. What is DHRUVA, and how does DIGIPIN relate to it?

Ans: DHRUVA (Digital Hub for Reference & Unique Virtual Address) is the DPI addressing ecosystem; DIGIPIN is its foundational layer.

8. Which characters from the beta version were replaced, and by what?

Ans: G, W, X were replaced by C, F, T.

9. What is the DIGIPIN of Dak Bhawan, New Delhi?

Ans: 39J49LL8T4 (displayed as 39J 49LL 8T4).

10. Name the five key components of the DHRUVA institutional framework.

Ans: Central Mapper (CM), AAVA, AIP, AIU, and AIA.

■ SOURCES & CURRENCY

Verified against: (1) PIB Press Release, Ministry of Communications, 27 May 2025; (2) the final **DIGIPIN Technical Document**, Department of Posts (March 2025); (3) the **DHRUVA** policy document (May 2025); (4) the official India Post DIGIPIN page and FAQ (indiapost.gov.in/digipin, reviewed April 2026); and (5) the official open-source repository (github.com/INDIAPOST-gov/digipin, Apache 2.0). The labelling grid and Dak Bhawan example were independently verified by encoding/decoding. Content reflects the position as on the date of preparation. As DIGIPIN/DHRUVA is an evolving DPI, always confirm the latest parameters from official India Post sources before relying on them for the examination.

— End of Booklet —

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